

Jane Street Quant Trader Internship — Full Topic List & Detailed Prep Plan

One honest note up front: no plan guarantees an offer — this process has a very low acceptance rate and interviewers are explicitly looking for how you think, not a checklist you've memorized. What *does* move your odds a lot is coverage (not having gaps) + repetition under real time pressure + practicing out loud with another person. This plan is built around those three levers.

PART 1 — COMPLETE TOPIC LIST BY SUBJECT

A. Mental Math

- Multiplication/division of 2-digit $\times$ 2-digit, 2-digit $\times$ 3-digit numbers
- Percentages (of a number, percentage change, reverse percentages)
- Fractions $\leftrightarrow$ decimals $\leftrightarrow$ percentages, quick simplification
- Squares/cubes of numbers up to ~ 30 , square roots by estimation
- Quick addition/subtraction chains (running totals, as in a P&L)
- Averaging, weighted averages
- Basic estimation/rounding tricks to sanity-check an answer fast

B. Combinatorics & Counting

- Permutations vs. combinations, when to use each
- Counting with restrictions (no two adjacent, at least one of X, exactly k of n)
- Stars and bars (distributing indistinguishable items into bins)
- Inclusion-exclusion principle
- Pigeonhole principle
- Counting paths on a grid / lattice paths

C. Core Probability

- Sample spaces, events, independence vs. mutual exclusivity
- Conditional probability and Bayes' theorem
- Law of total probability
- Discrete distributions: binomial, geometric, hypergeometric, Poisson, negative binomial
- Continuous distributions: uniform, exponential, normal (basic properties, not deep calculus)
- Expected value and linearity of expectation (the single most-used trick in these interviews)
- Variance, standard deviation, covariance (conceptual, not heavy computation)
- Order statistics (probability the max/min of n draws exceeds some value)

- Geometric probability (e.g., "two people agree to meet between 12 and 1, each waits 15 min — probability they meet")

D. Stochastic / Sequential Reasoning

- Random walks (gambler's ruin, probability of reaching a boundary first)
- Markov chains (basic transition reasoning, absorbing states)
- Martingale concept (why doubling-down strategies don't beat a fair game in expectation)
- Optional stopping / stopping-time intuition (conceptual level — know why you can't "beat" a fair coin by choosing when to stop)
- Streak problems (probability of a run of k consecutive heads in n flips)

E. Decision Theory / Betting & Risk

- Expected value vs. expected utility (why maximizing EV isn't always the right personal decision)
- Kelly criterion — derivation intuition and when to bet less than "full Kelly"
- Risk-neutral vs. risk-averse framing
- Variance/bankroll management in repeated betting
- Basic game theory: dominant strategies, zero-sum reasoning, simple auctions (why sealed-bid auctions incentivize certain bidding behavior)

F. Market-Making & Trading Intuition

- What a bid-ask spread compensates a market maker for
- Adverse selection (why you widen your quote when you suspect the other side knows more than you)
- Inventory risk and skewing your quote based on position
- Updating a "fair value" estimate as new information arrives (Bayesian updating in real time)
- Basic auction mechanics (first-price vs second-price/Vickrey intuition)
- Position sizing and stop-loss thinking in a game context
- How to behave in a multi-round game where your score carries over (don't blow up your book on round 1)

G. Logic Puzzles & Brainteasers (non-probability)

- Weighing/balance puzzles (find the odd coin in n weighings)
- Symmetry arguments to shortcut a problem
- Induction-style puzzles (prove a pattern holds for n , extend from $n-1$)
- Sequence/pattern recognition
- Classic riddles used as warm-ups (these test composure more than math)
- Fermi estimation (order-of-magnitude questions — "how many piano tuners in NYC"-style reasoning)

H. Programming (general, language-agnostic)

- Writing clean logic from scratch without autocomplete
- Basic data structures (arrays, hash maps, sets) and when to use each
- Simple simulation scripts (e.g., Monte Carlo simulate a probability puzzle to check your analytical answer)
- Comfort with Python specifically (used in the Algorithmic Trading elective): pandas/numpy basics, writing a simple strategy loop against simulated data
- Debugging out loud / talking through logic errors calmly

I. Communication & Behavioral

- Narrating your reasoning as you go, not just at the end
 - How to respond productively to a hint or correction mid-problem
 - Explaining any project on your resume in plain, non-jargon language
 - Concrete stories: a time you were wrong and changed your mind; a time you worked well on a team; what draws you to trading/Jane Street specifically
 - Questions to ask your interviewers (genuine ones — about their day-to-day, how teams are structured, etc.)
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PART 2 — DETAILED WEEK-BY-WEEK PLAN (12 weeks, ~1-1.5 hrs/day)

Compress to 6 weeks by merging two weeks into one if your timeline is short — don't skip categories, just shorten volume.

Weeks 1-2: Foundations (Topics A, B, C)

- Daily: 15 min Zetamac (track score; log it in a spreadsheet)
- Daily: 1-2 combinatorics/counting problems, written out fully
- Read Jane Street's Probability and Markets Guide cover to cover; redo every worked example without looking
- Start *Fifty Challenging Problems in Probability* (Mosteller) — 2-3 problems every other day
- Learn/re-derive: linearity of expectation, Bayes' theorem, binomial/geometric/Poisson basics

Weeks 3-4: Depth in probability + start decision theory (Topics C, D, E)

- Continue Zetamac daily, now timed harder (aim for your target pass rate)
- Move to *Heard on The Street* (Crack) — one timed puzzle/day (5-10 min cap), written solution
- Learn random walks / gambler's ruin, Markov chain basics, martingale intuition
- Read a short primer on Kelly criterion; solve 3-4 bet-sizing problems
- Start narrating solutions out loud, alone, recorded on your phone — listen back once a week

Weeks 5-6: Market-making intuition + logic puzzles (Topics F, G)

- Learn bid-ask spread / adverse selection / inventory skew conceptually (a few short articles or the Jane Street guide's markets section is enough — no textbook needed)
- Start informal trading games with a friend: they hold a hidden number/range, you make markets and adjust as they reveal info
- Daily brainteaser/logic puzzle from a curated bank (see resources) — mix probability and non-probability puzzles
- Weekly: first real mock interview with a friend — one probability puzzle, solved fully out loud, with them interrupting to give hints/pushback

Weeks 7-8: Programming + Fermi estimation + more mocks (Topics H, plus G continued)

- Write 2-3 short Python scripts simulating puzzles you've solved analytically (Monte Carlo check) — this cements intuition AND builds coding comfort
- Practice 3-4 Fermi/estimation problems, timed, out loud
- Weekly mock interview, now adding a market-making game component
- Start a running list of every mistake you make in practice — review it weekly, not just once

Weeks 9-10: Full-loop simulation (all topics combined)

- Twice this fortnight: simulate a full Superday with a friend — probability puzzle → market-making game → brainteaser → 2 behavioral questions, back to back, no break, like the real thing
- Refine resume talking points — be able to explain any project in under 90 seconds, in plain language
- Write and rehearse your 2-3 behavioral stories (situation → what you did → what you'd do differently)
- Keep Zetamac and daily puzzle rotation going, lighter volume now (maintenance, not building)

Weeks 11-12: Taper and polish

- Focus remaining mock sessions specifically on your weakest category from the mistake log
- One last full-loop mock in week 11
- Week 12: light review only — re-skim your mistake log, redo 2-3 of your favorite puzzles for confidence, then rest. Cramming new material in the final 48 hours tends to hurt clarity more than it helps.

PART 3 — WHAT ACTUALLY MOVES THE ODDS MOST (ranked)

1. **Weekly live mock interviews with a partner**, not solo study — this is the single highest-leverage activity, because the skill being graded (reasoning out loud, adapting to hints, staying composed) only trains in a live setting.

2. **Volume of puzzles solved fully out loud**, timed — quantity with the right conditions beats quantity done silently on paper.
3. **A mistake log you actually revisit** — most people solve a puzzle once, feel good, and move on. Revisiting *why* you got something wrong (rushed? misread the question? wrong framework?) compounds much faster than fresh problems.
4. **Mental math to a solid floor** — doesn't need to be elite, but if it's a bottleneck it can end the process early regardless of everything else.
5. Everything else (breadth of topics, programming, behavioral polish) — necessary but lower-leverage than the above four.

PART 4 — RESOURCE LIST

Resource	Use for	Link
Jane Street Probability and Markets Guide	Official primer, topics C–F	https://www.janestreet.com/probability-markets/
Jane Street Puzzles archive	Monthly puzzles, similar flavor	https://www.janestreet.com/puzzles/
Jane Street careers/interviewing hub	Process overview, mock video	https://www.janestreet.com/join-jane-street/
Zetamac	Mental math drills (Topic A)	https://arithmetic.zetamac.com
<i>Fifty Challenging Problems in Probability</i> — Mosteller	Core probability puzzles (C)	print/library/booksellers
<i>Heard on The Street</i> — Crack	Classic quant interview puzzles (C, D, E)	print/booksellers
<i>A First Course in Probability</i> — Ross	Filling theory gaps (C, D)	standard textbook, library/booksellers
Tradermath brainteaser database	Curated, filterable practice puzzles (C, G)	https://www.tradermath.org/brainteasers

PART 5 — WEEKLY SELF-CHECK

At the end of each week, honestly rate yourself 1–5 on:

- ☐ Mental math speed/accuracy
- ☐ Probability puzzle-solving speed

- ☐ Clarity when narrating out loud
- ☐ Composure when given a hint or told you're wrong
- ☐ Market-making/trading-game instincts
- ☐ Behavioral story readiness

Anything sitting at 2 or below going into week 9 is where your remaining mock sessions should concentrate.